

HCI PROJECT

MicroMove

Task, Storyboard & Prototypes

Final presentation of the subject Fondamenti di Comunicazione e internet.

TEAM MEMBER

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Politecnico di Milano

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VALUE PROPOSITION

Focus: Student physical well-being at the Politecnico di Milano Campus.

**The challenge —> Engineering students face long hours of sedentary study leading to:
Physical strain, Mental fatigue, Awareness Gap...**

Goal: Integrating micro-activity into the academic routine without disrupting concentration



RESTORE ENERGY



REDUCE
DISCOMFORT



WELL-TIMED



CURRENT SITUATION

The Problem



Sedentary Behaviour

Uninterrupted sitting sessions often exceeding 90-120 minutes.



Time Blindness

Hyper-focus leads to losing track of time and ignoring the body's warning signs



Physical Toll

Accumulation of stiffness in the neck, shoulders, and lower back.

I only realize i'm in pain when i finally try to stand up.

"User Insight



MICROMOVE APPROACH

The Solution



Discreet Detection



Context-Aware

Suggests invisible or socially acceptable micro-breaks suitable for libraries.



Seamless Integration

Supports behavior change without interrupting concentration or attracting attention.



The Silent Constraint: Social Inhibition

_ I feel like everyone is watching me
if I start stretching my arms."

CURRENT SITUATION

The Problem



Social Judgment

Students avoid movement to maintain the study etiquette of shared spaces



Disturbance

Fear of distracting others with noise or large gestures



The result

Students choose physical discomfort over social awkwardness



MICROMOVE APPROACH

The Solution



MicroMove's Mission:

Breaking this barrier by making movement **invisible** and **socially acceptable**.

Research Methodology: Deepening User Understanding

Phase 1: Qualitative Interviews (Needfinding)



- 4 in-depth interviews with engineering students
 - Diverse profiles (from 'Extreme Active' to 'Extreme Sedentary')
- Goal: understand **motivations**, **routines**, and **barriers**.

Phase 2: Contextual Inquiry (Refinement)



- **Direct observation** in **study environments** (library, labs)
 - Focus: posture, breaks, and social behavior.

Key Research Insight

Discovered a Social Barrier: Students feel embarrassed to stretch or move in public study areas, prioritizing academic momentum over physical comfort.

Identifying the User: Personas & Scenarios

Ruben

21, Computer Engineering



Represents Time Blindness.
Needs: Simplicity and immediate triggers.

Elena

22, Industrial Engineering



Represents Social Inhibition.
Needs: Discretion and context-aware suggestions.

Core User Needs



Context-awareness

Adapts to user's location and activity.



Efficiency

Quick, effortless interactions to save time.



Discretion

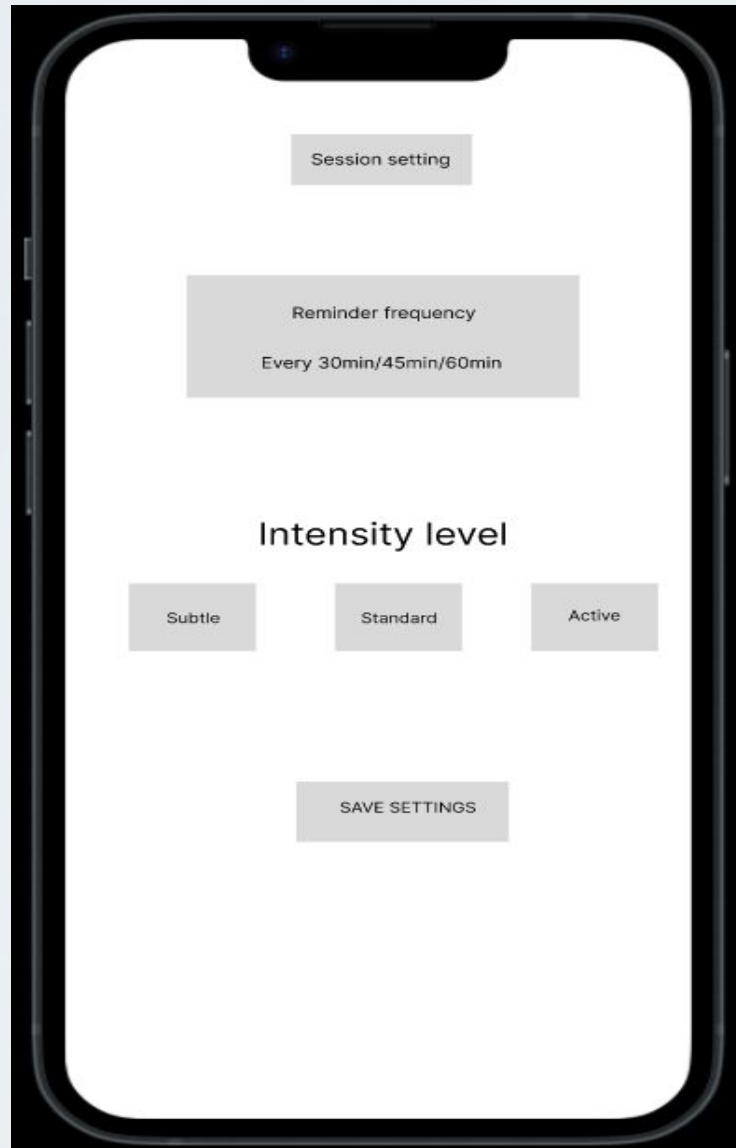
Subtle and private notifications.

Scenario: Discreet Micro-stretch at the Library



Elena receives a discreet haptic alert on her device in the library. She performs a subtle, context-aware micro-stretch, avoiding drawing attention.

UX Research: Posture Management Solutions - User Feedback



Option 1: Mobile App

Pros:

- High familiarity for most users.
- Better for detailed data visualization (stats, trends, progress tracking).

Cons:

- Can be visually distracting in social or professional settings.
- Requires taking the phone out in quiet or restricted zones (e.g., meetings, libraries).



Option 2: Smartwatch (Wearable)

- Pros:
Excellent for haptic feedback (vibrations); very discreet for libraries; supports 2-second micro-interactions.
- Cons:
Limited screen space for complex navigation.

Core Functionalities: Task Analysis

1. Simple Task



Responding to a Reminder

Goal: React to a haptic alert and confirm a 30-second neck stretch.

2. Moderate Task



Session Configuration

Goal: Set a 2-hour study block with 'Library Mode' (silent, vibration-only alerts).

3. Complex Task

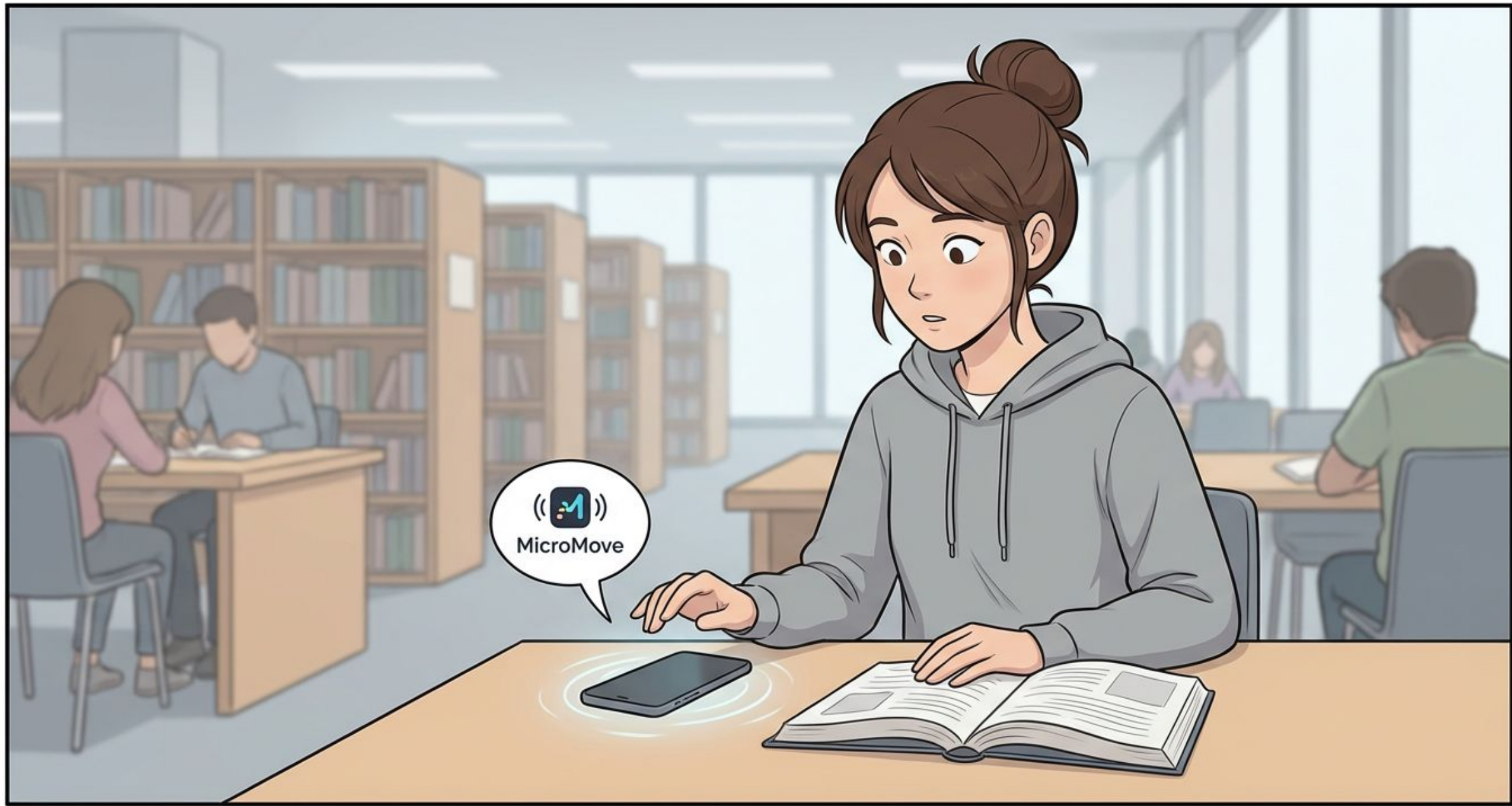


Progress Management

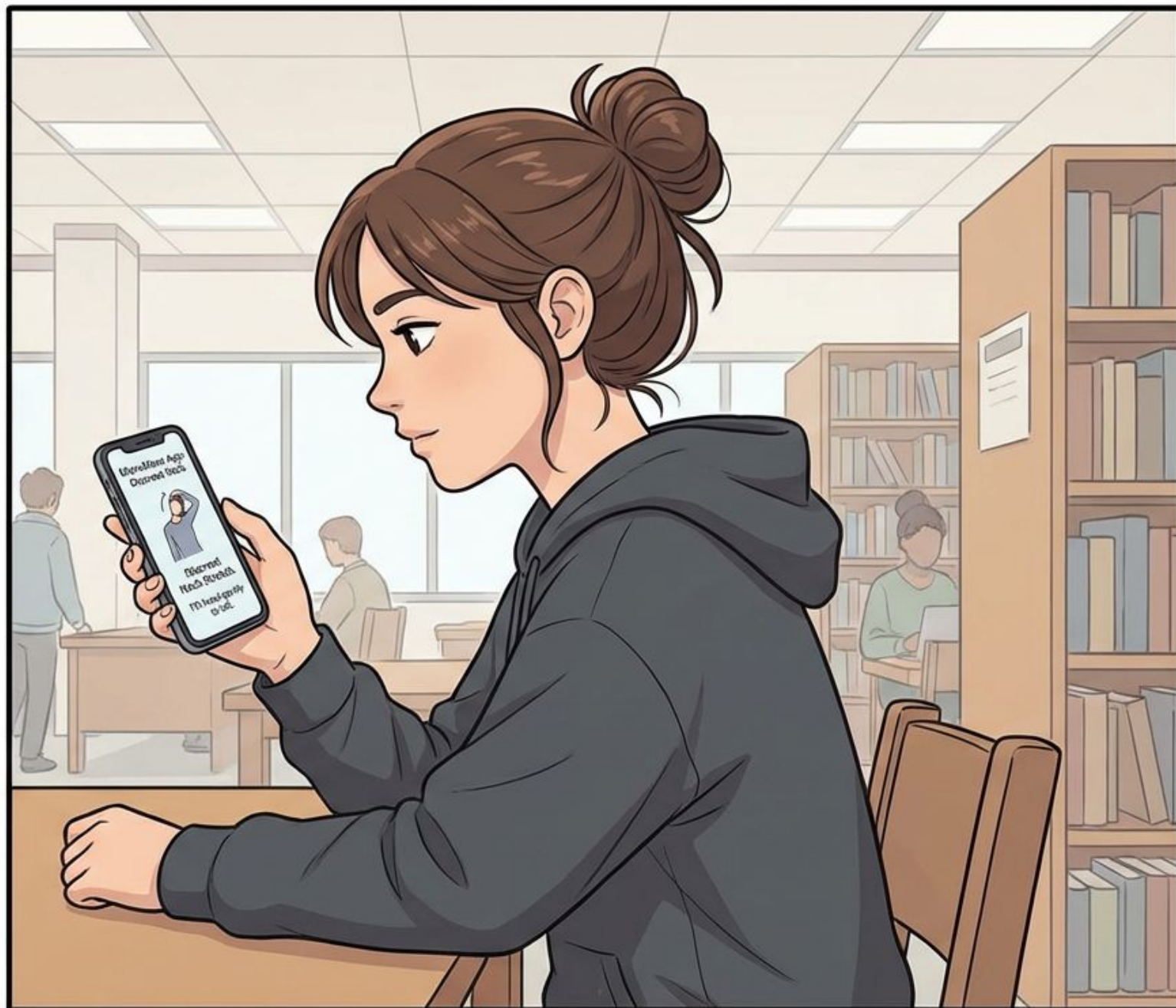
Goal: Navigate to the history dashboard to review 'Active Minutes' and long-term trends.



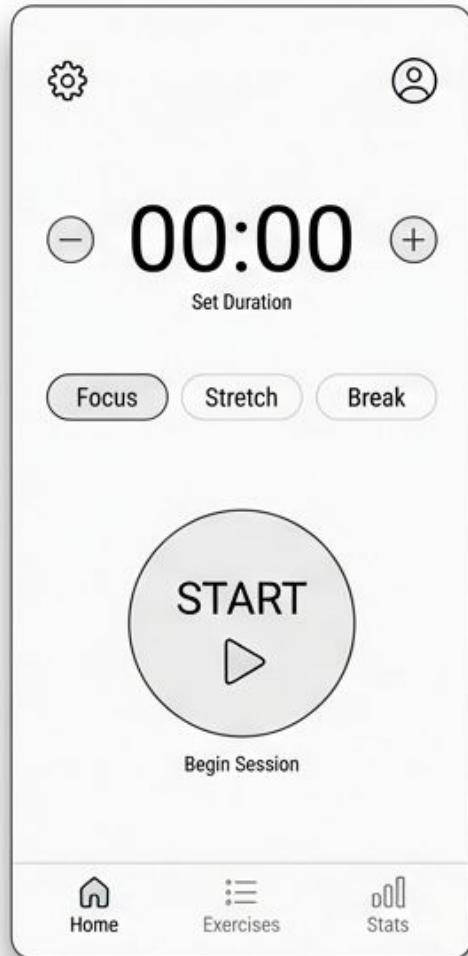
PANEL 1: ELENA STARTS A DEEP STUDY SESSION.



BCT LIBRARY



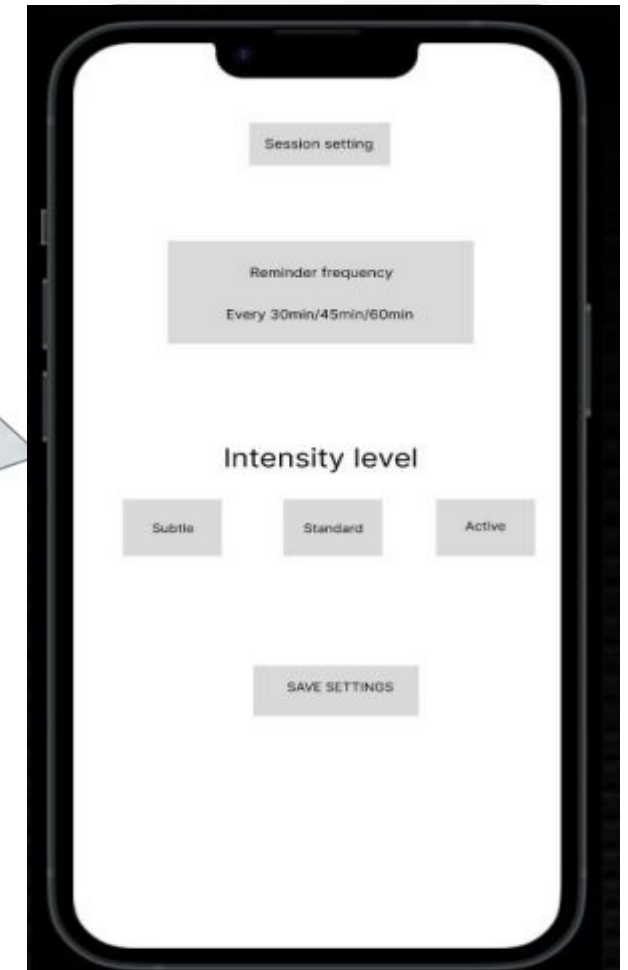
TIMER SCREEN



EXERCISES SCREEN



PROGRESS/STATS SCREEN



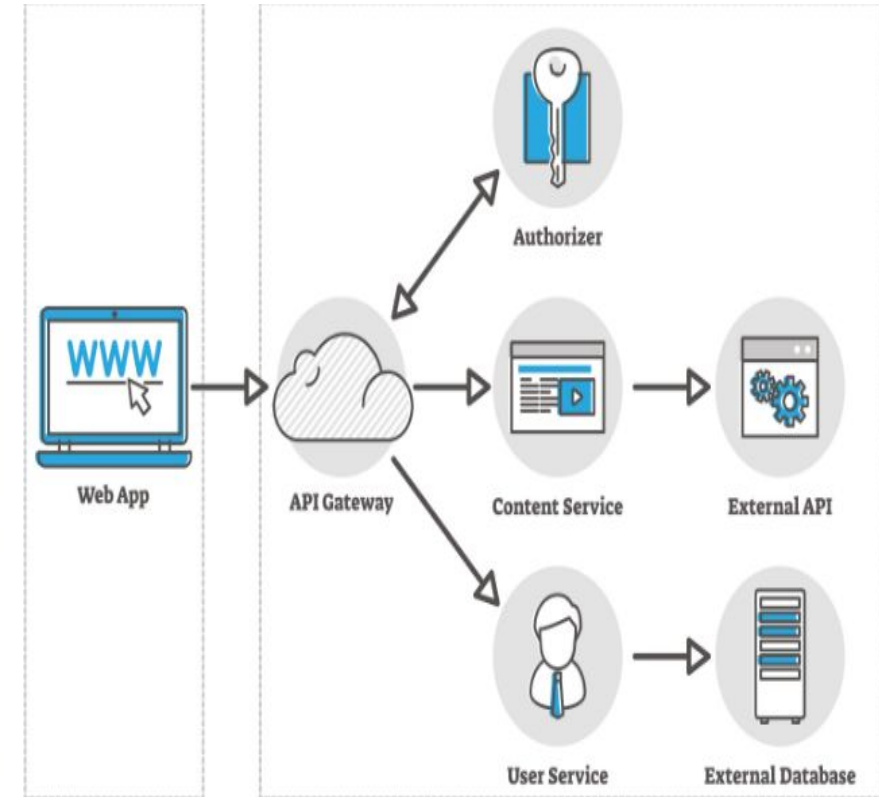
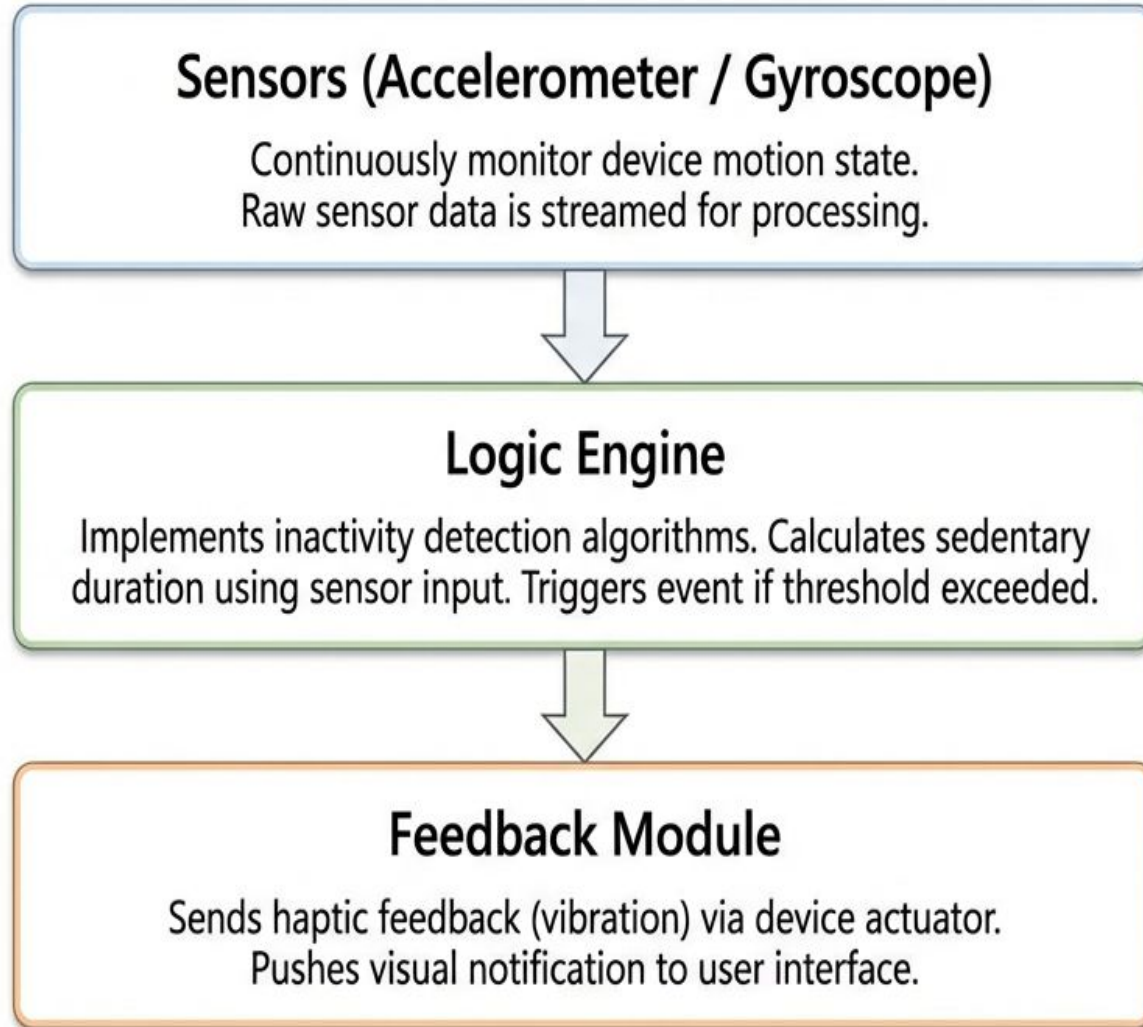
User-Centric Navigation: A hierarchical structure where the 'Start' action is always prominent to minimize friction.

Visual Simplicity: Use of large, recognizable icons for exercises to reduce cognitive load—students can understand the movement at a glance without reading.

Clean Layout: Minimalist design intentionally crafted to avoid being another source of distraction during study sessions.



Technical Feasibility & Architecture

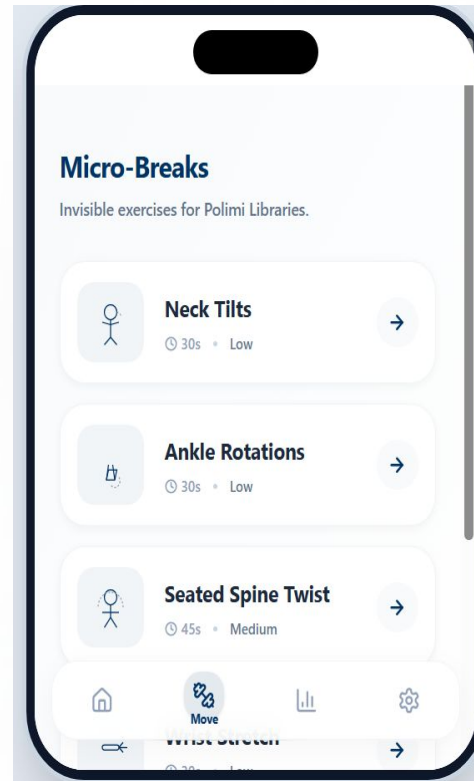


USER TESTING & RESULTS

✓ Success Rate: 100%

★ Satisfaction: 4.7/5

🕒 Efficiency: <20s



- 1) "Library Mode" praised for social comfort.
- 2) Visual clarity of icons enabled immediate action.
- 3) "Polimi Blue" aesthetics felt professional and familiar.

Final Refinements & Future Steps

Final Improvements

- **Ergonomics:** Increased 'Start' button size for better touch target.
- **Accessibility:** Improved contrast for library lighting.
- **UX Writing:** Changed 'Stats' to 'My Progress' for a personal touch.

Future Vision

- Integration with Smartwatches for haptic-only alerts.
- Gamified 'Polimi Wellness' challenges.

MicroMove proves that we can balance academic productivity with physical well-being. Thank you for your attention!